

# Breakfast Cereals

## Data

These data come from the 1993 ASA Statistical Graphics Exposition, and are taken from the mandatory FDA food labels. The data have been normalized for a portion of one American cup. The original data are available at <http://lib.stat.cmu.edu/datasets/1993.expo/>.

You can load the dataset (called **cereal**) using the following command:

```
download.file("http://stat.duke.edu/~cr173/Sta102_Fa14/Proj/cereal.Rdata", method="wget", destfile="cereal.Rdata")
load("cereal.Rdata")
```

Below is a description of the variables:

1. **mfr**: Manufacturer, represented by its first initial: G=General Mills, K=Kelloggs, N=Nabisco, P=Post, Q=Quaker Oats, R=Ralston Purina.
2. **calories**: number of calories in one portion.
3. **protein**: grams of protein in one portion.
4. **fat**: grams of fat in one portion.
5. **sodium**: milligrams of sodium in one portion.
6. **fibre**: grams of dietary fibre in one portion.
7. **carbo**: grams of complex carbohydrates in one portion.
8. **sugars**: grams of sugars in one portion.
9. **shelf**: display shelf (1, 2, or 3, counting from the floor).
10. **potassium**: grams of potassium.
11. **vitamins**: vitamins and minerals (none, enriched, or 100%).

## Research Question

Is there a significant difference between the calories and fat content of cereals placed on the different shelves? Based on these results can you say if healthier cereals tend to be displayed on the upper or lower shelves?